

Graph Puzzles III.3-preview: Oriented five-cycle double covers of snarks with colouring defect three

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ABSTRACT: For every optimal triple of perfect matchings of a snark of colouring defect three, the triple's hexagonal core extends to an oriented 5-cycle double cover. We construct the five directed cycle members. Three auxiliary orientations of the bichromatic exterior paths determine an ordered pair of cover labels on every edge. At each cubic vertex the three ordered pairs form a directed triangle. Consequently every label induces a directed even subgraph, while every edge belongs to exactly two such subgraphs with opposite directions.

1 An ordered-pair criterion

Here a *cycle* means an even subgraph and may be disconnected. We use *snark* for a connected bridgeless cubic graph with no proper 3-edge-colouring. A *3-array* is a triple of perfect matchings. The multiplicity of an edge is the number of matchings of the triple that contain it; the *core* of the triple consists of the edges whose multiplicity is not one. The *colouring defect* of a cubic graph is the minimum, over all 3-arrays, of the number of edges of multiplicity zero.

Definition 1.1. An *oriented 5-cycle double cover* of a graph G is an indexed family $(D_1, \dots, D_5)$ of directed cycles such that every edge belongs to exactly two members and receives opposite directions in those two members.

Lemma 1.2 (Ordered-pair criterion). *Give every edge a reference direction and an ordered pair $p \rightarrow q$ of distinct labels from $\{1, \dots, 5\}$. Let D_p use the edge in its reference direction and let D_q use it in the reverse direction. Suppose that, when the three incident edges at each vertex are read outwards—retaining $p \rightarrow q$ if the reference direction points out of the vertex and reversing it otherwise—their ordered pairs form a directed triangle. Then $(D_1, \dots, D_5)$ is an oriented 5-cycle double cover.*

Proof. Each edge lies in precisely the two members named by its pair, with opposite directions. At a vertex, every label occurring in the directed triangle occurs once as a tail and once as a head. Hence each D_r has one edge entering and one leaving the vertex, or no incident edge there. Thus every D_r is a directed even subgraph. $\square$

2 The defect-three construction

Theorem 2.1. *Let G be a snark with colouring defect 3. For every optimal triple of perfect matchings of G , there is an oriented 5-cycle double cover having the triple's induced hexagonal core as one indexed member.*

Proof. Fix an arbitrary triple of perfect matchings attaining the colouring defect. By the defect-three core characterization [KMNS24]¹, Theorem 3.3, their core is an induced cycle

$$C = v_1 v_2 \cdots v_6 v_1.$$

Every edge of $G - E(C)$ is singly covered, and membership in the three matchings gives a proper 3-edge-colouring of $G - E(C)$. After rotating C and permuting the colours, the six boundary edges at $v_1, \dots, v_6$ have colours

$$1, 1, 2, 2, 3, 3. \tag{1}$$

Let F_i denote the exterior colour- i class.

Three auxiliary orientations

Fix $i \in \{1, 2, 3\}$, write $\{i, j, k\} = \{1, 2, 3\}$, and put

$$P_i = F_j \cup F_k, \quad K_i = C \cup P_i.$$

Every vertex outside C has degree two in P_i . Exactly four vertices of C are incident with P_i , so P_i consists of two paths pairing these four terminals, together with disjoint circuits. Suppressing every degree-two vertex in the component of K_i containing C produces a cubic multigraph Q_i consisting of a 4-cycle and a perfect matching.

We claim that Q_i is bipartite. By rotating and relabelling, it is enough to take $i = 1$. The four terminals are v_3, v_4, v_5, v_6 , in that

¹ Ján Karabáš, Edita Máčajová, Roman Nedela and Martin Škoviera. *Cubic graphs with colouring defect 3*. Electronic Journal of Combinatorics 31(2) (2024), P2.6. doi:10.37236/12333.

cyclic order. A 4-cycle plus a perfect matching is non-bipartite only when the matching pairs the opposite vertices v_3v_5 and v_4v_6 . If this occurred, interchange colours 2 and 3 on the exterior Kempe path joining v_3 to v_5 . The boundary word would become

$$1, 1, 3, 2, 2, 3.$$

This colouring extends across C : give the consecutive cycle edges v_rv_{r+1} , with indices modulo 6, the colours

$$3, 2, 1, 3, 1, 2,$$

respectively. At every v_r , the boundary edge and the two cycle edges then receive all three colours. This would Tait-colour G , a contradiction. Hence every Q_i is bipartite.

Choose a bipartition $A_i \cup B_i$ of Q_i . Orient its four suppressed C -segments from B_i to A_i , and orient its two matching edges from A_i to B_i . Lift these directions consistently through the suppressed paths, and orient every circuit component of P_i cyclically. Call the resulting orientation of K_i O_i .

Thus O_i passes through every degree-two vertex. At a core vertex whose boundary edge has colour i , its two C -edges point in opposite directions. At either endpoint of a P_i -path, the two C -edges both point in and the exterior edge points out, or all three directions are reversed.

The five cover labels

Fix an arbitrary reference direction on every edge. For $e \in K_i$, put $\sigma_i(e) = +$ or $-$ according as O_i agrees or disagrees with the reference direction; for $e \in F_i$, put $\sigma_i(e) = 0$. The sign word

$$\sigma(e) = (\sigma_1(e), \sigma_2(e), \sigma_3(e))$$

assigns an ordered pair to e by the following table:

edge class	displayed sign words	ordered pairs	
C	$+++$, $+-$, $-+-$, $---$	$1 \rightarrow 5, 2 \rightarrow 5, 3 \rightarrow 5, 4 \rightarrow 5$	(2)
F_1	$0++$, $0+-$	$1 \rightarrow 2, 3 \rightarrow 4$	
F_2	$+0+$, $+0-$	$1 \rightarrow 3, 2 \rightarrow 4$	
F_3	$++0$, $+-0$	$1 \rightarrow 4, 2 \rightarrow 3$	

The negative of a displayed sign word receives the reversed ordered pair. Use the pair $p \rightarrow q$ in lemma 1.2: the member D_p takes the reference direction and D_q takes the reverse direction. In the vector calculations below we identify $+$ and $-$ with $+1$ and -1 . At an end-vertex v of an edge e , let $\sigma_v(e) = \sigma(e)$ if the reference direction points

away from v , and let $\sigma_v(e) = -\sigma(e)$ otherwise. Thus $\sigma_v(e)$ is the sign word read outwards at v , and its table pair is reversed in the second case.

It remains only to check the local directed-triangle condition. Put

$$\begin{aligned} t_1 &= (1, 1, 1), & t_2 &= (1, -1, -1), \\ t_3 &= (-1, 1, -1), & t_4 &= (-1, -1, 1). \end{aligned}$$

The four vectors $t_1, \dots, t_4$ are affinely independent. The first row of (2) says that a core sign word is t_p for $p \rightarrow 5$, or $-t_p$ for $5 \rightarrow p$. The remaining rows say that an exterior sign word belonging to the pair $p \rightarrow q$ is

$$\frac{t_p - t_q}{2}.$$

First let $v \notin V(C)$. For each coordinate i , the orientation O_i passes through v on the two edges whose colours differ from i . Their two outward i -signs are therefore opposite, while the colour- i edge has sign 0. Hence the three vectors $\sigma_v(e)$ sum to zero. If their outward-read ordered pairs are $p_s \rightarrow q_s$, then

$$\sum_{s=1}^3 (t_{p_s} - t_{q_s}) = 0.$$

For $r \in \{1, 2, 3, 4\}$, let d_r be the number of these pairs with tail r , minus the number with head r . The displayed equality says $\sum_r d_r t_r = 0$, while automatically $\sum_r d_r = 0$. Affine independence gives $d_r = 0$ for every r . Thus every label occurs equally often as a tail and as a head. Three non-loop ordered pairs with this property form a directed triangle.

Now let $v \in V(C)$, and let its exterior edge a have colour i . Suppose that the outward-read table pair of a is $p \rightarrow q$. Then

$$\sigma_v(a) = \frac{t_p - t_q}{2}, \quad (t_p)_i = (t_q)_i, \quad (t_p)_j = -(t_q)_j \quad (j \neq i).$$

In O_i , the two C -edges have opposite signs. In each O_j with $j \neq i$, they have the same sign, opposite to that of a . Consequently their two sign words are t_q and $-t_p$. The table labels the two C -edges by $q \rightarrow 5$ and $5 \rightarrow p$. Together with $p \rightarrow q$, they form the directed triangle

$$p \longrightarrow q \longrightarrow 5 \longrightarrow p.$$

The hypothesis of lemma 1.2 now holds at every vertex. Therefore $(D_1, \dots, D_5)$ is an oriented 5-cycle double cover. By the table, label 5 occurs precisely on the core edges. Hence $E(D_5) = E(C)$, and D_5 is a cyclic orientation of C .

Finally, all five members are nonempty. Indeed, if some D_r were empty, all edge-pairs would lie in the complete graph on the remaining four labels. Colour each edge of G by the unique one of the three one-factors of this K_4 containing its pair. At every vertex the three pairs form a triangle and hence receive the three distinct colours. This would Tait-colour G , again a contradiction. $\square$

References

- [KMNS24] Ján Karabáš, Edita Máčajová, Roman Nedela and Martin Škoviera. *Cubic graphs with colouring defect 3*. Electronic Journal of Combinatorics 31(2) (2024), P2.6. doi:[10.37236/12333](https://doi.org/10.37236/12333).